

CIRROSE NA UTI - 1



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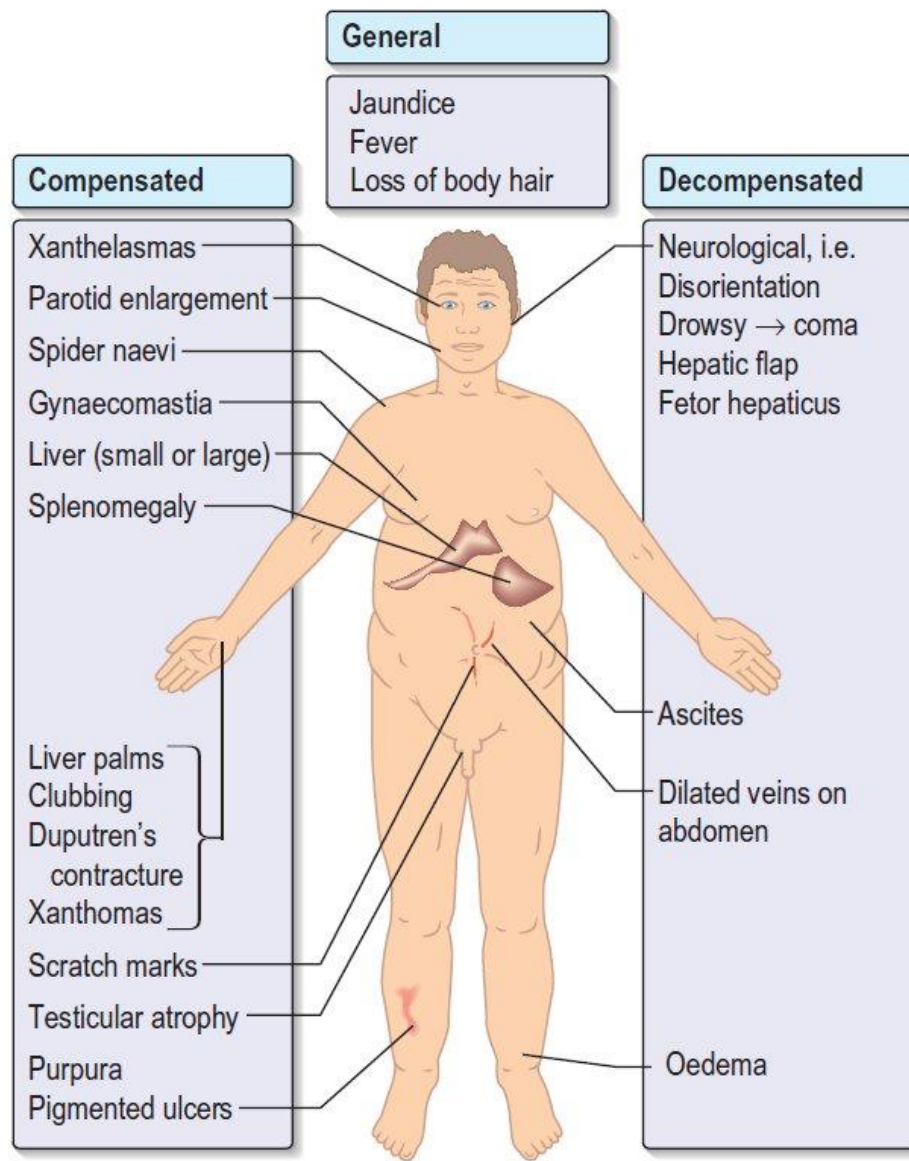


1 CUIDADOS NEUROLÓGICOS

2 HEMODINÂMICA PARTE 1



DOENÇA SISTÊMICA



Physical signs in chronic liver disease.

<https://www.grepmed.com/images/3550/cirrhosis-signs-symptoms-diagram-diagnosis>



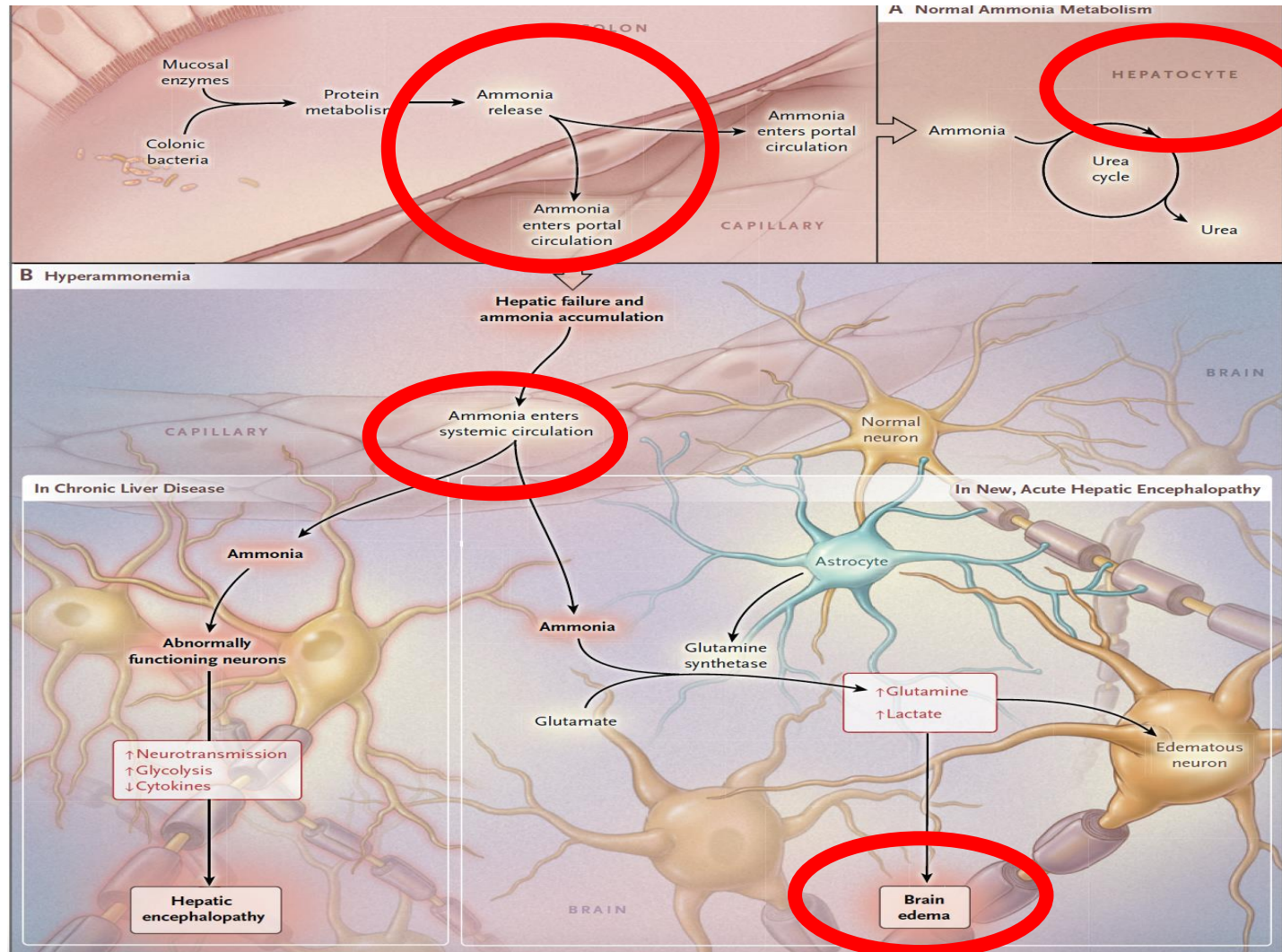
ENCEFALOPATIA HEPÁTICA X HIC



Hepatic Encephalopathy

Author: Eelco F.M. Wijdicks, M.D., Ph.D. [Author Info & Affiliations](#)

Published October 27, 2016 | N Engl J Med 2016;375:1660-1670 | DOI: 10.1056/NEJMra1600561 | VOL. 375 NO. 17



Shunt /TIPS ?



Aumentar eliminação no cólon:

Laxativos

Modificar flora cólon:

Lactulose / Rifaximina

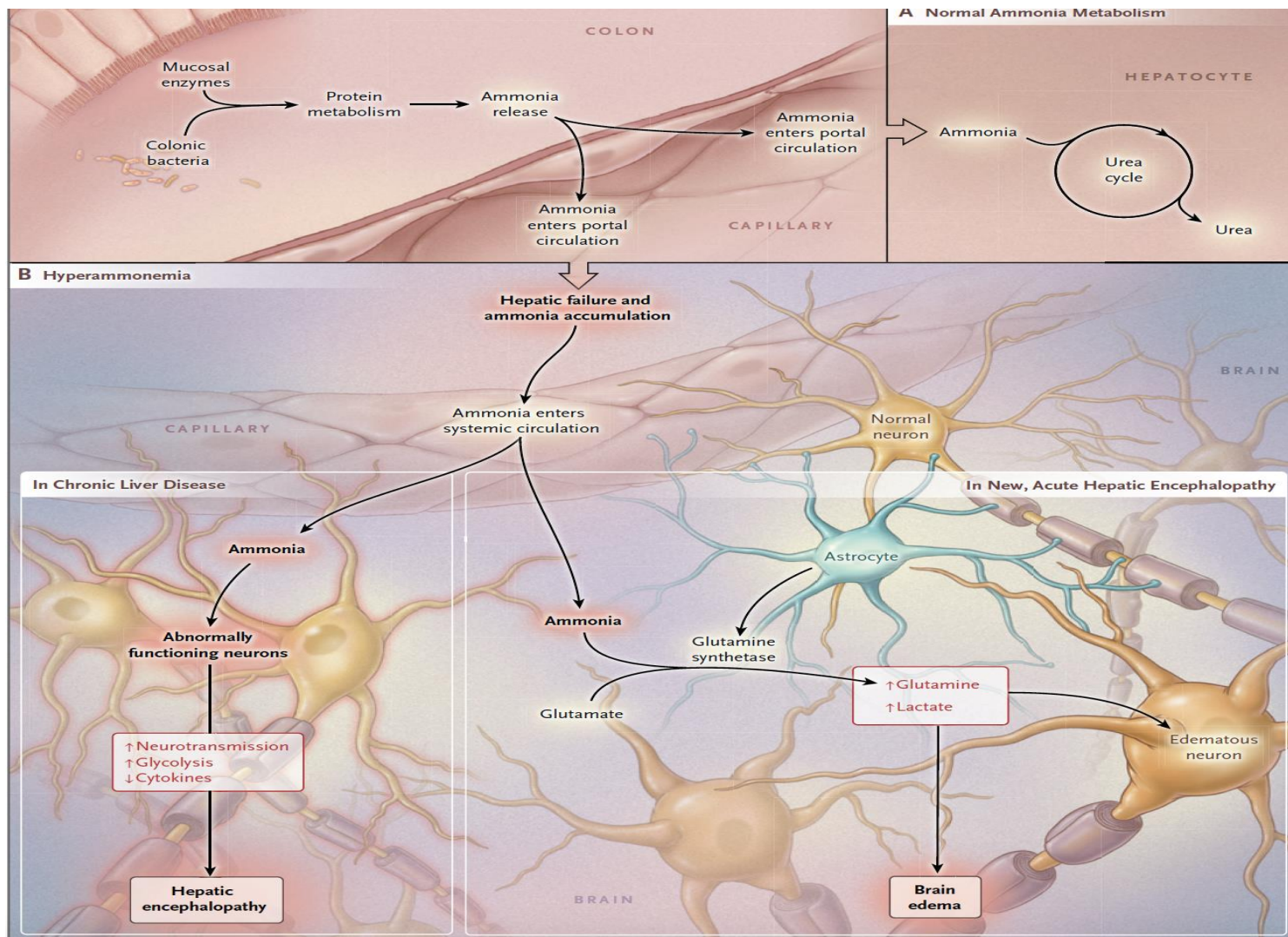
REVIEW ARTICLE

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Hepatic Encephalopathy

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Ciclo da Ureia >> Lo-La

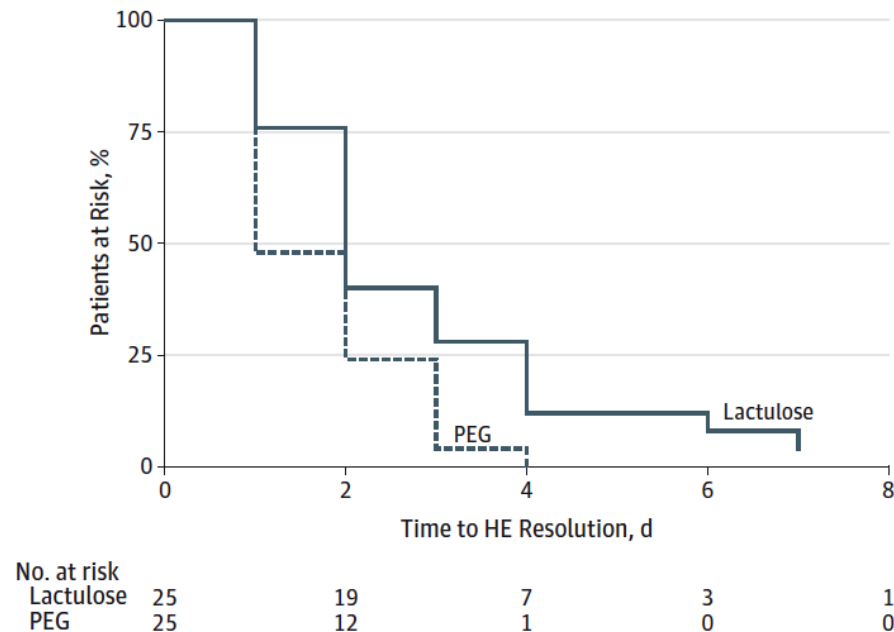
Aumento de clearance sérico:
Hemofiltração contínua



Lactulose vs Polyethylene Glycol 3350-Electrolyte Solution for Treatment of Overt Hepatic Encephalopathy The HELP Randomized Clinical Trial

Robert S. Rahimi, MD, MS; Amit G. Singal, MD, MS; Jennifer A. Cuthbert, MD; Don C. Rockey, MD

Figure 3. Time to Hepatic Encephalopathy (HE) Resolution



Shown is a Kaplan-Meier graph depicting the proportion of patients at risk for HE who received either standard-of-care lactulose therapy or polyethylene glycol 3350-electrolyte solution (PEG). Patients receiving PEG had more rapid resolution of HE than those receiving lactulose ($P = .01$).

UTI X AMBULATÓRIO

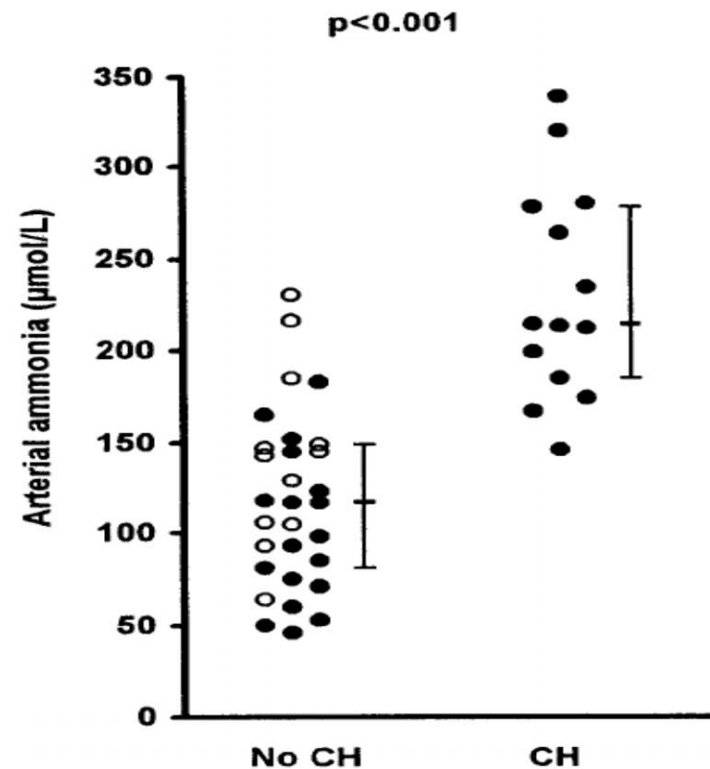
- ❖ Não necessariamente precisa ser sempre lactulose!
- ❖ Não necessariamente retirar Diuréticos
- ❖ Dissacarídeo x problemas
- ❖ Disfunção intestinal ?



CUIDADOS NEUROLÓGICOS

Cerebral Herniation in Patients With Acute Liver Failure Is Correlated With Arterial Ammonia Concentration

JENS OTTO CLEMMESSEN, FIN STOLZE LARSEN, JENS KONDRUP, BENT ADEL HANSEN, AND PETER OTT



CRÔNICO X AMÔNIA

ANNALS OF **Hepatology**

ORIGINAL ARTICLE

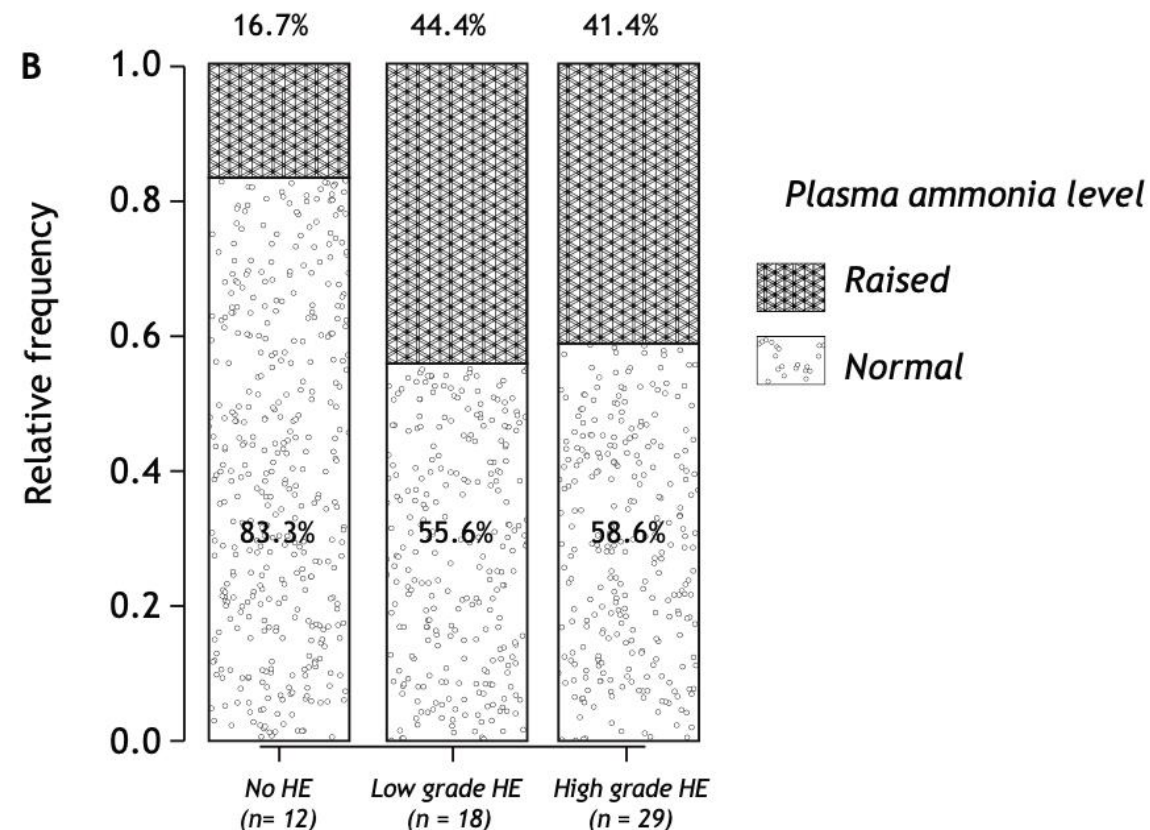
January-February, Vol. 12 No.1, 2013: 108-114

How to diagnose hepatic encephalopathy in the emergency department

Felix Gundling,* Edin Zelihić,** Holger Seidl,* Bernhard Haller,***
Andreas Umgelter,**** Wolfgang Schepp,* Christoph Dodt**

- EH PODE TER AMÔNIA < 65
- SEM EH NÃO COSTUMA > 65

- Diagnostic accuracy of blood ammonia was 59.3% [95% CI = 45.7 to 71.9%; sensitivity 47.2% (95% CI = 30.4 to 64.5%), specificity 78.3% (95% CI = 56.3% to 92.5%)].



ORIGINAL RESEARCH ARTICLE

Comparison of Arterial vs Venous Ammonia Levels in Hepatic Encephalopathy

Nisha Kajla

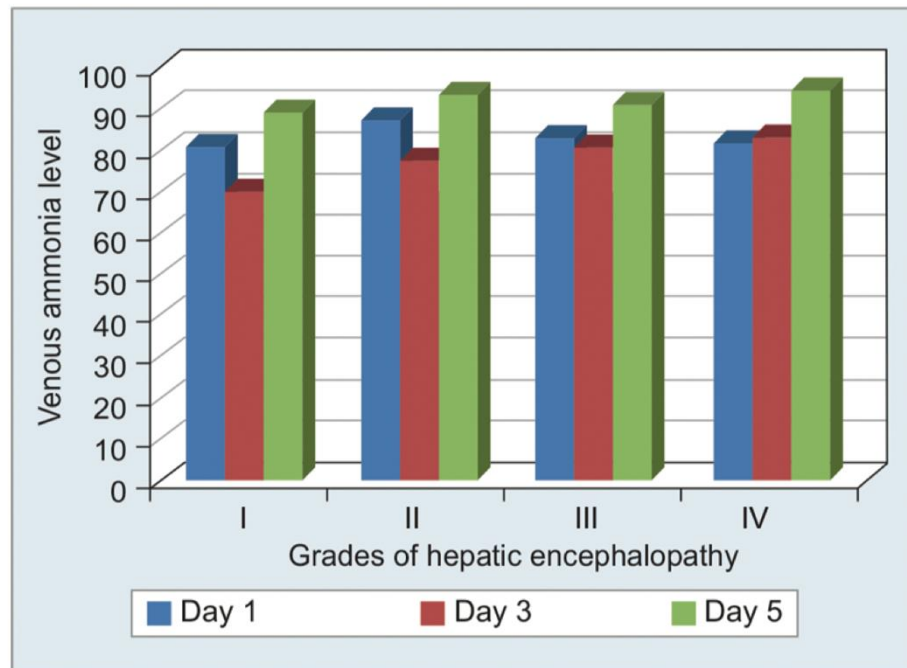


Fig. 7: Correlation of venous ammonia levels with West Haven grading of encephalopathy

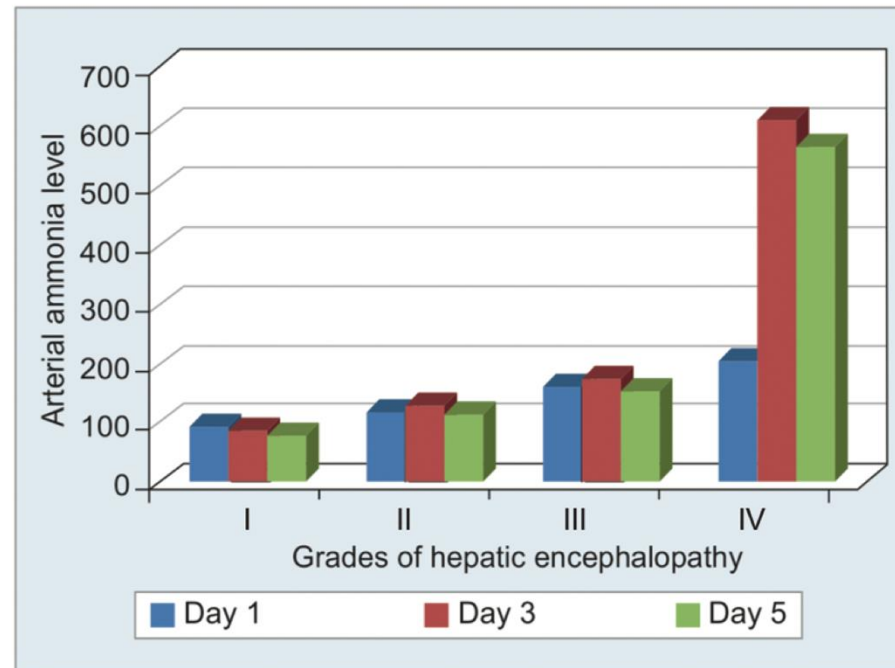


Fig. 6: Correlation of arterial ammonia levels with West Haven grading of encephalopathy

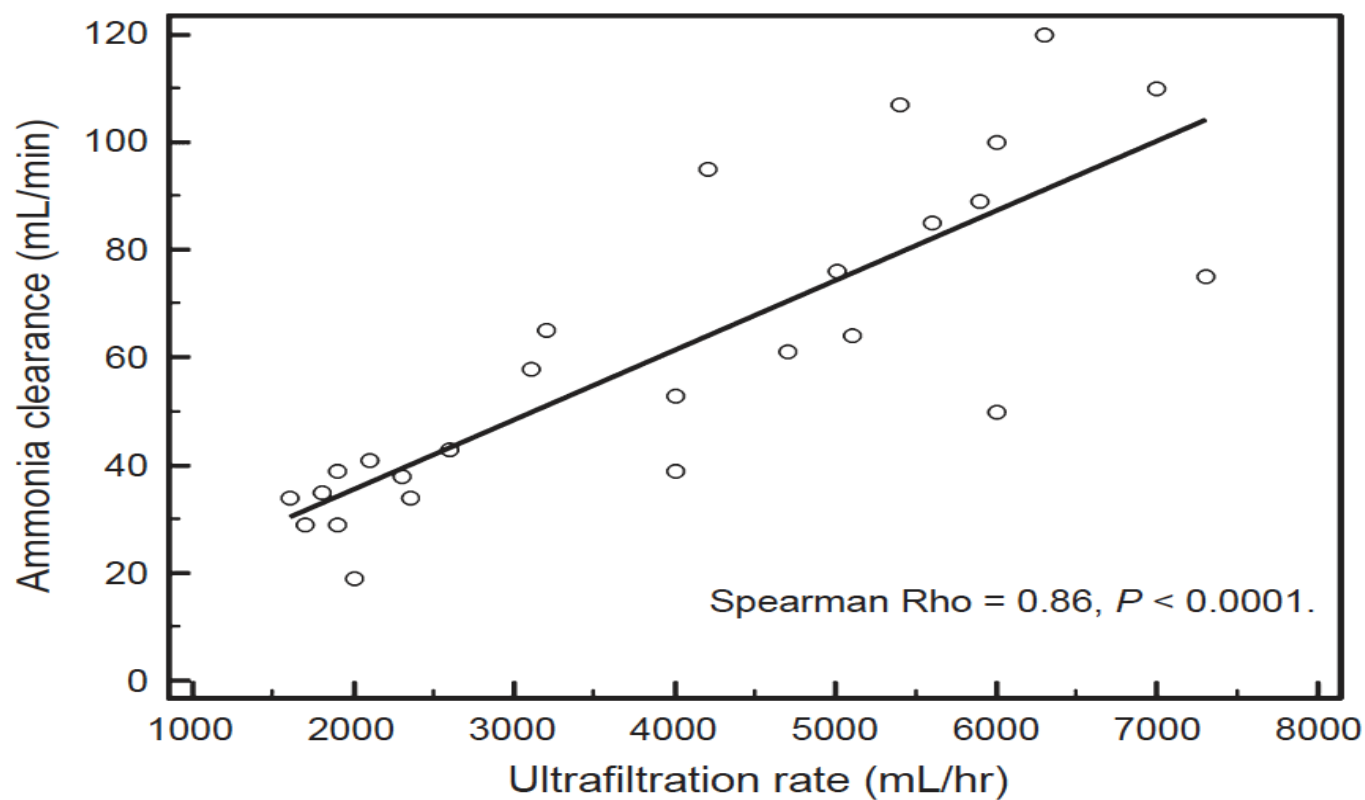


CIRRHOSIS AND LIVER FAILURE

Ammonia clearance with haemofiltration in adults with liver disease

Andrew J. Slack, Georg Auzinger, Chris Willars, Tracy Dew, Rebecca Musto, Daniel Corsilli, Roy Sherwood, Julia A. Wendon and William Bernal

Institute of Liver Studies, King's College Hospital Foundation Trust, London, UK



HEMODINÂMICA



Infection and Predictors of Outcome of Cirrhotic Patients after Emergency Care Hospital Admission

Wajima Safi,* Mayada Elnegouly,^{†,‡} Raphael Schellnegger,[§]
Katrin Umgelter,^{||} Fabian Geisler,[†] Wolfgang Reindl,[¶] Bernd Saugel,** Alexander Hapfelmeier,^{††} Andreas Umgelter^{†,‡}

This study enrolled all consecutive patients who presented to the non-operative Emergency Department (ED)

Patients with infection

In 80 (36.5%) patients [54 (68%) male] a clinical diagnosis of infection was established within the first 48 h. All



Intestinal Permeability in Cirrhotic Patients With and Without Spontaneous Bacterial Peritonitis: Is the Ring Closed?

Emidio Scarpellini, MD¹, Venanzio Valenza, MD², Maurizio Gabrielli, MD¹, Ernesto Cristiano Lauritano, MD¹, Germano Perotti, MD², Giuseppe Merra, MD¹, Antonio Dal Lago, MD¹, Veronica Ojetti, MD¹, Maria Elena Ainora, MD¹, Michele Santoro, MD¹, Giovanni Ghirlanda, MD¹ and Antonio Gasbarrini, MD¹

Am J Gastroenterol 2010; 105:323–327; doi:10.1038/ajg.2009.558; published online 20 October 2009

METHODS: IP was evaluated by the ⁵¹Cr-ethylenediaminetetraacetic acid (⁵¹Cr-EDTA) permeability test in 52 LC patients and in 48 sex- and age-matched controls. The presence of ⁵¹Cr-EDTA was also evaluated in ascitic fluid after therapeutic paracentesis in all LC patients with ascites.

RESULTS: An altered IP was found in 45% of LC patients compared with 4% of controls ($P<0.00001$). vs. 50% of those without SBP). ⁵¹Cr-EDTA was present in all ascitic samples obtained from patients with SBP compared with 22% of patients without SBP.



CIRROSE X SEPSE ?

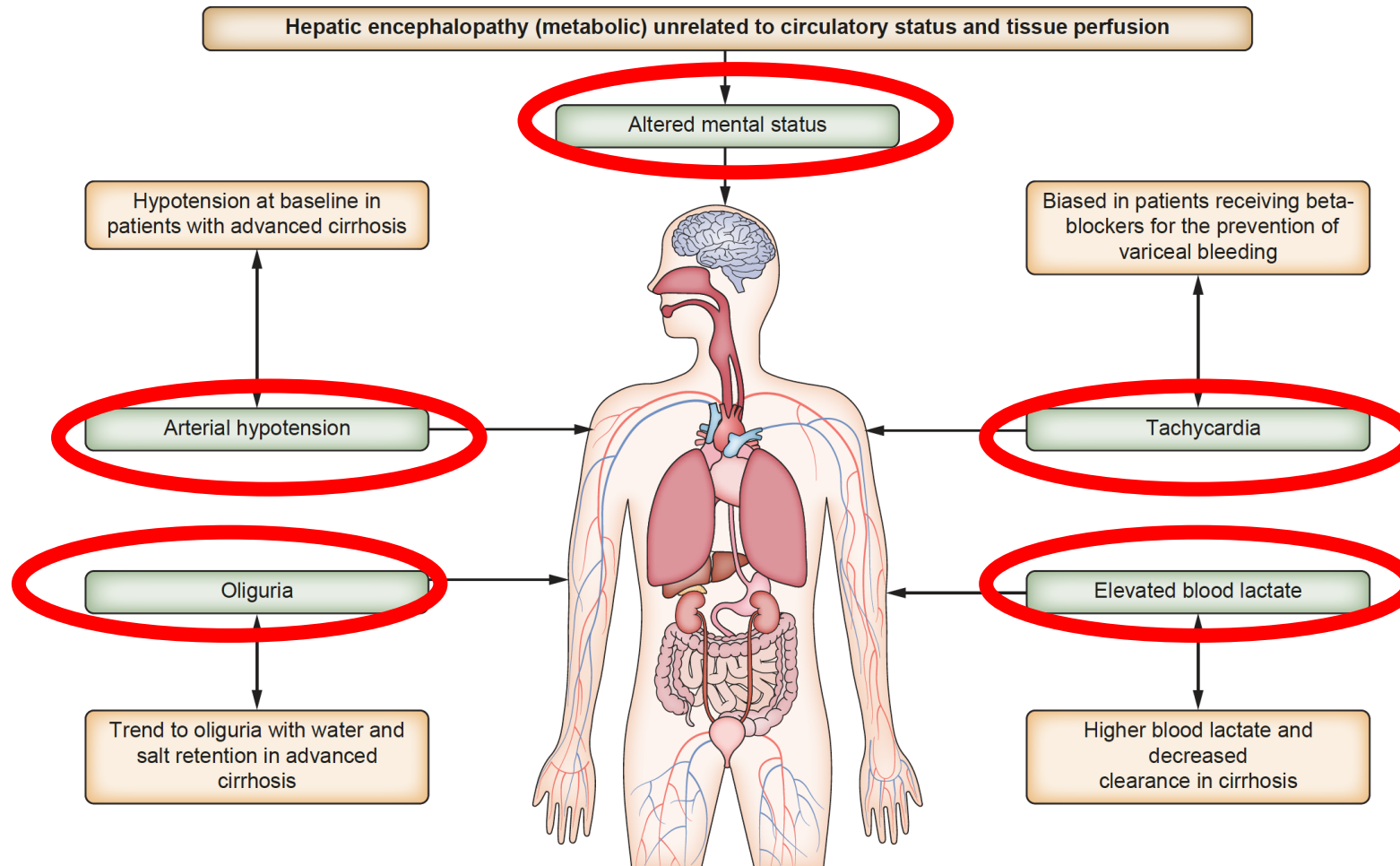


Fig. 1. Initial assessment of hypovolemia and/or shock in the general population and potential pitfalls in patients with cirrhosis.





COMO GUIAR ?

FLUIDO



VASOPRESSOR



ANTIBIÓTICO



EASL Clinical Practice Guidelines for the management of patients with decompensated cirrhosis[☆]

European Association for the Study of the Liver*

HEMODINÂMICA

Hipovolemia relativa

- Aumento DC x vasodilatação

Cardiomiopatia da cirrose

One study in 173 transplant recipients assessed systolic (resting ejection fraction <55%) and diastolic (E/A ratio <1 or a deceleration time >200 ms) dysfunction and reported it occurring in 2% and 43% of patients, respectively. Whilst patients with

study followed 80 patients to assess one-year mortality, finding 46% had diastolic dysfunction on echo criteria and about half of these had grade II dysfunction, in whom mean arterial blood

Tips/Tx



Cardiomiopatia da Cirrose

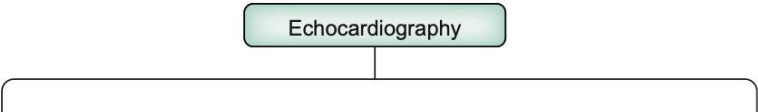


Table 2. Prevalence of CCM according to the different diagnostic criteria.

Authors (Patients)	2005 WGC	2019 CCC	p value
Razpotnik <i>et al.</i> ⁶¹ (consecutive patients with cirrhosis)	66.2%	55.7% with GLS>22% 19.7% with GLS<18% only	p = n.s. p <0.05
Singh <i>et al.</i> ²⁵ (LT candidates)	74.8%	85.6% with GLS<18% only	p = n.s.
Spann <i>et al.</i> ⁴¹ (LT candidates)	77%	30% without GLS	p <0.05
Arman <i>et al.</i> ⁶² (LT recipients)	78.6%	17.6%	p <0.01
Cesari <i>et al.</i> ²⁴ (consecutive patients with cirrhosis)		29% including GLS>22%	
Izzy <i>et al.</i> ²³ (LT recipients)		38.8%	

CCC, Cirrhotic Cardiomyopathy Consortium; GLS, global longitudinal strain; LT, liver transplant; WGC, World Congress of Gastroenterology.

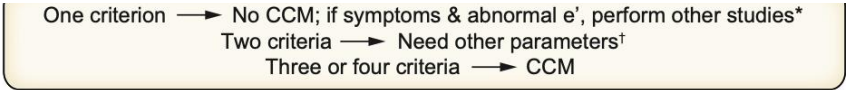


Fig. 2. Diagnosis of cirrhotic cardiomyopathy using echocardiography. *Diastolic stress echocardiography or right heart catheterisation. †This includes the ratio of pulmonary vein peak systolic velocity to peak diastolic velocity or systolic time velocity integral to diastolic time-velocity integral, isovolemic relaxation time, left atrial strain, or left ventricular mass index and relative wall thickness. CCM, cirrhotic cardiomyopathy; GLS, global longitudinal strain; LAVI, left atrial volume index; LVEF, left ventricular ejection fraction; TR, tricuspid regurgitation.



Exploration of skin perfusion in cirrhotic patients with septic shock

Arnaud Galbois^{1,2,3,*}, Naïke Bigé^{1,2}, Claire Pichereau^{1,2}, Pierre-Yves Boëlle^{2,4,5}, Jean-Luc Baudel¹, Simon Bourcier^{1,2}, Eric Maury^{1,2,5}, Bertrand Guidet^{1,2,5}, Hafid Ait-Oufella^{1,2,6}

TEC

?

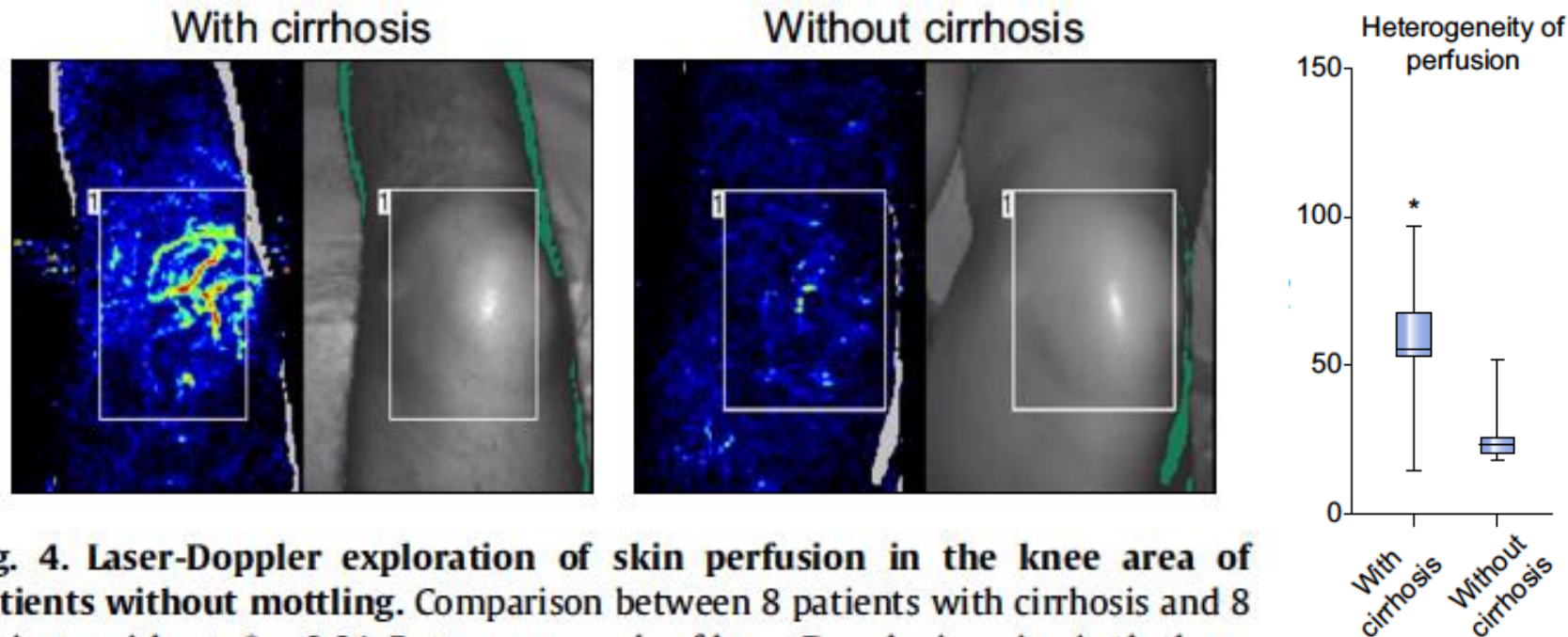


Fig. 4. Laser-Doppler exploration of skin perfusion in the knee area of patients without mottling. Comparison between 8 patients with cirrhosis and 8 patients without: $*p < 0.01$. Bottom: example of laser-Doppler imaging in the knee

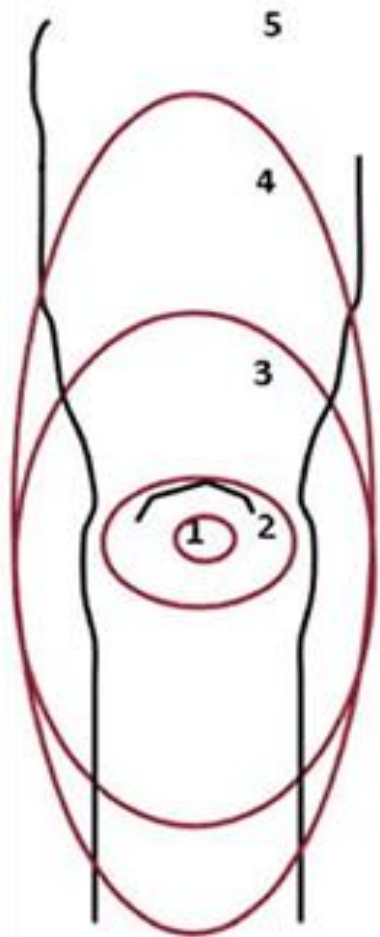
skin perfusion was longer preserved in patients with cirrhosis;
perfusion is higher in the patient with cirrhosis.



Exploration of skin perfusion in cirrhotic patients with septic shock

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Simon Bourcier^{1,2}, Eric Maury^{1,2,5}, Bertrand Guidet^{1,2,5}, Hafid Ait-Oufella^{1,2,6}

MOTTLING SCORE



Mottling score



Stage 2



Stage 3



Stage 5



Exploration of skin perfusion in cirrhotic patients with septic shock

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MACRO X MICRO

Macrocirculação

Table 2. Hemodynamic parameters at H6 in patients with liver cirrhosis according to their survival status at day 14: univariate analysis.

Parameters:	OR (95% CI)	p value
MAP (mmHg)		
≥65	1	
<65	2.1 (0.4-11.8)	0.46
Heart rate (beats/min)		
>100	1	
≤100	3.0 (0.7-12.2)	0.17
Central venous pressure (mmHg)		
≥8	1	
<8	1.2 (0.1-13.1)	1
Cardiac output (L/min)		
≥5.5	1	
<5.5	1.3 (0.3-5.2)	1
Urinary output (ml/kg/h)		
≥0.5	1	
<0.5	3.8 (0.8-15.9)	0.09

Microcirculação

Arterial lactate (mmol/L)		
<5	1	
≥5	5.2 (1.2-23.3)	0.04
Lactate clearance (%)		
>0	1	
≤0	8.0 (1.8-35.8)	0.009
ScvO ₂ (%)		
≥70	1	
<70	15.0 (0.8-280.7)	0.08
Mottling score		
≤1	1	
>1	42.4 (2.3-785.9)	0.0001

MAP, mean arterial pressure; ScvO₂, central venous oxygen saturation.

Liver cirrhosis affects serum lactate level measurement while assessing disease severity in patients with sepsis

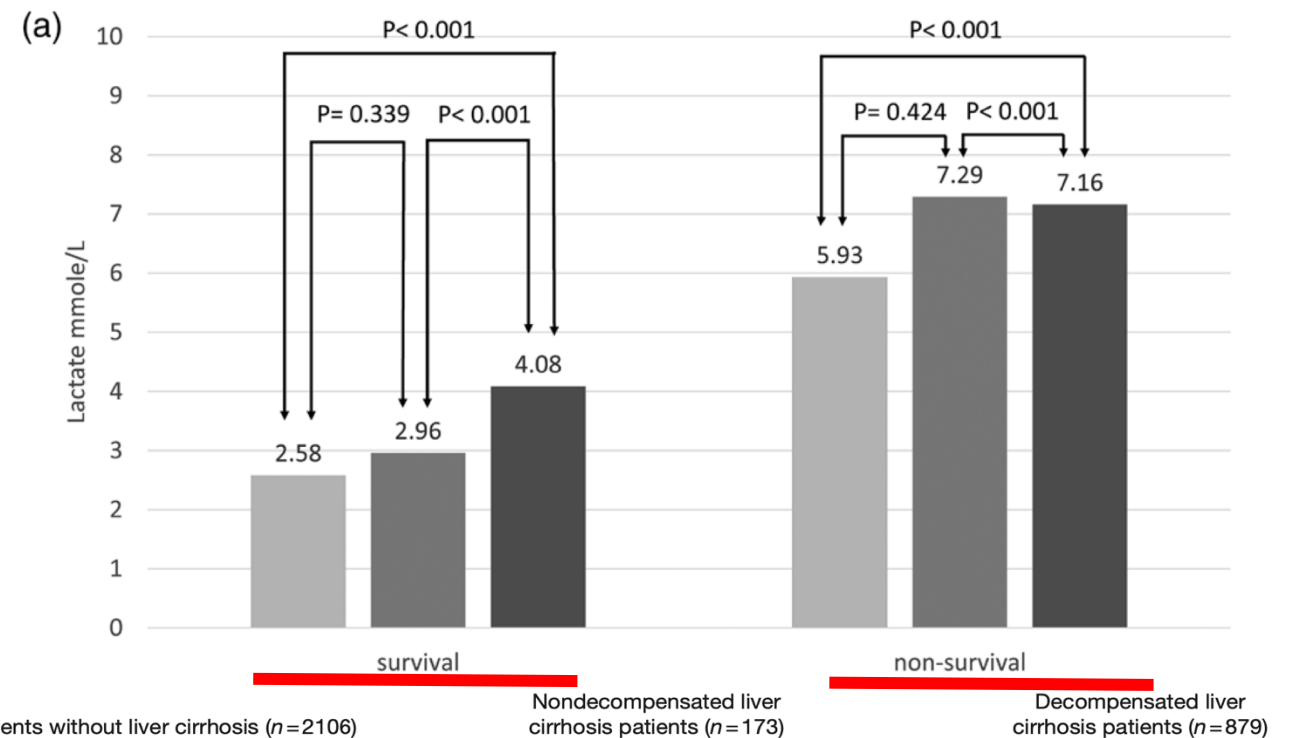
Chi-Yung Cheng^{a,b}, Chia-Te Kung^a, Kuan-Han Wu^a, Fu-Cheng Chen^a, Hsien-Hung Cheng^a, Fu-Jen Cheng^a, Jyun-Bin Huang^a and Chih-Min Su^{a,c}

European Journal of Gastroenterology & Hepatology 2021, 33:1201–1208

Lactato

Cirrhosis affects lactate level in sepsis Cheng *et al.*

www.eurojgh.com



Coletado em: 12/10/2020 17:25

LACTATO, SANGUE ARTERIAL

LACTATO >270 mg/dL 4,5 a 14,4 mg/dL Eletrodo ion seletivo

14/10/2020 08:41:00

Pedido : 20200100443990

Coletado em: 14/10/2020 08:40

LACTATO, SANGUE ARTERIAL

LACTATO 16 mg/dL 4,5 a 14,4 mg/dL

Exame realizado no Urgências Laboratoriais - Centro Cirurgico

LACTATO ?



Predicting stroke volume and arterial pressure fluid responsiveness in liver cirrhosis patients using dynamic preload variables

A prospective study of diagnostic accuracy

Chun-Yu Wu, Ya-Jung Cheng, Ying-Ju Liu, Tsung-Ta Wu, Chiang-Ting Chien and Kuang-Cheng Chan, on behalf of the NTUH Center of Microcirculation Medical Research (NCMMR)

CIRROSE X VOLUME

Table 2 Changes in haemodynamic parameters before and after fluid loading in stroke volume fluid responders and non-responders

SV responders (n = 23)		SV non-responders (n = 39)	
Before	After	Before	After

62%

Table 3 Changes in haemodynamic parameters before and after fluid loading in mean arterial pressure fluid responders and non-responders

Arterial pressure responders (n = 16)		Arterial pressure non-responders (n = 46)	
Before	After	Before	After

74%



5. For patients with sepsis induced hypoperfusion or septic shock we **suggest** that at least 30 mL/kg of intravenous (IV) crystalloid fluid should be given within the first 3 h of resuscitation

Weak recommendation, low-quality evidence



GUIDELINES

Surviving sepsis campaign: international guidelines for management of sepsis and septic shock 2021



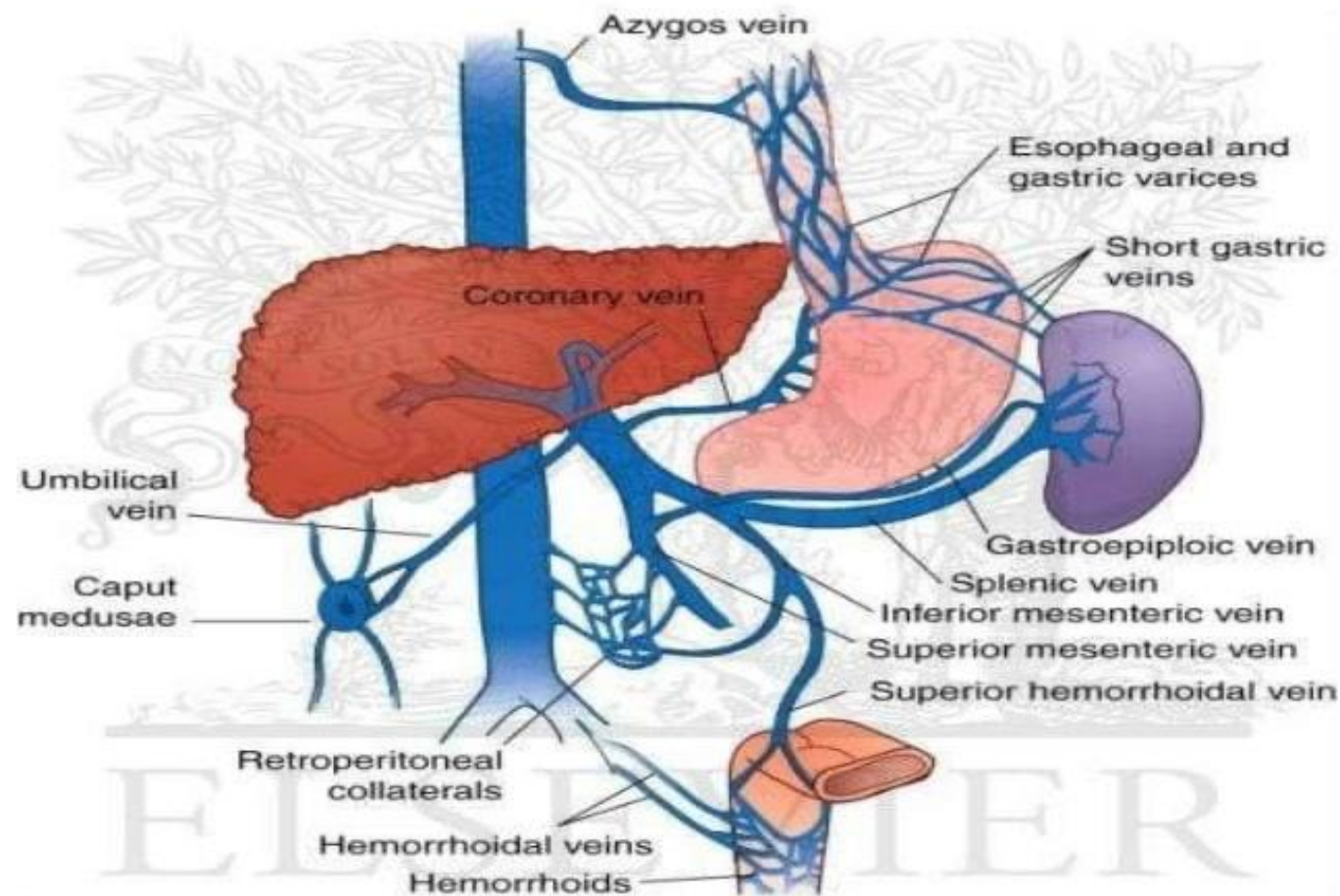
Laura Evans^{1*}, Andrew Rhodes², Waleed Alhazzani³, Massimo Antonelli⁴, Craig M. Coopersmith⁵, Craig French⁶, Flávia R. Machado⁷, Loralyn McIntyre⁸, Marlies Ostermann⁹, Hallie C. Prescott¹⁰, Christa Schorr¹¹, Steven Simpson¹², W. Joost Wiersinga¹³, Faye Alshamsi¹⁴, Derek C. Angus¹⁵, Yaseen Arabi¹⁶, Luciano Azevedo¹⁷, Richard Beale⁹, Gregory Beilman¹⁸, Emilie Belley-Cote¹⁹, Lisa Burry²⁰, Maurizio Cecconi^{21,22}, John Centofanti²³, Angel Coz Yataco²⁴, Jan De Waele²⁵, R. Phillip Dellinger¹¹, Kent Doi²⁶, Bin Du²⁷, Elisa Estenssoro²⁸, Ricard Ferrer²⁹, Charles Gomersall³⁰, Carol Hodgson³¹, Morten Hylander Møller³², Theodore Iwashyna³³, Shevin Jacob³⁴, Ruth Kleinpell³⁵, Michael Klompas^{36,37}, Younsuck Koh³⁸, Anand Kumar³⁹, Arthur Kwizera⁴⁰, Suzana Lobo⁴¹, Henry Masur⁴², Steven McLaughlin⁴³, Sangeeta Mehta⁴⁴, Yatin Mehta⁴⁵, Mervyn Mer⁴⁶, Mark Nunnally⁴⁷, Simon Oczkowski³, Tiffany Osborn⁴⁸, Elizabeth Papathanassoglou⁴⁹, Anders Perner⁵⁰, Michael Puskas⁵¹, Jason Roberts^{52,53,54,55}, William Schweickert⁵⁶, Maureen Seckel⁵⁷, Jonathan Sevransky⁵, Charles L. Sprung^{58,59}, Tobias Welte⁶⁰, Janice Zimmerman⁶¹ and Mitchell Levy⁶²

45 kg >> 1350ml/3h

70 Kg >> 2100ml/3h

100 kg >> 3000ml/3h

PRA ONDE O FLUIDO VAI ?



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<https://br.pinterest.com/sergiinikol/portal-hypertension/>



The portal pressure-blood volume relationship in cirrhosis¹

DAVID S. ZIMMON AND RICHARD E. KESSLER

From the Medical and Surgical Services, Veterans Administration Hospital, and New York University School of Medicine, New York, New York

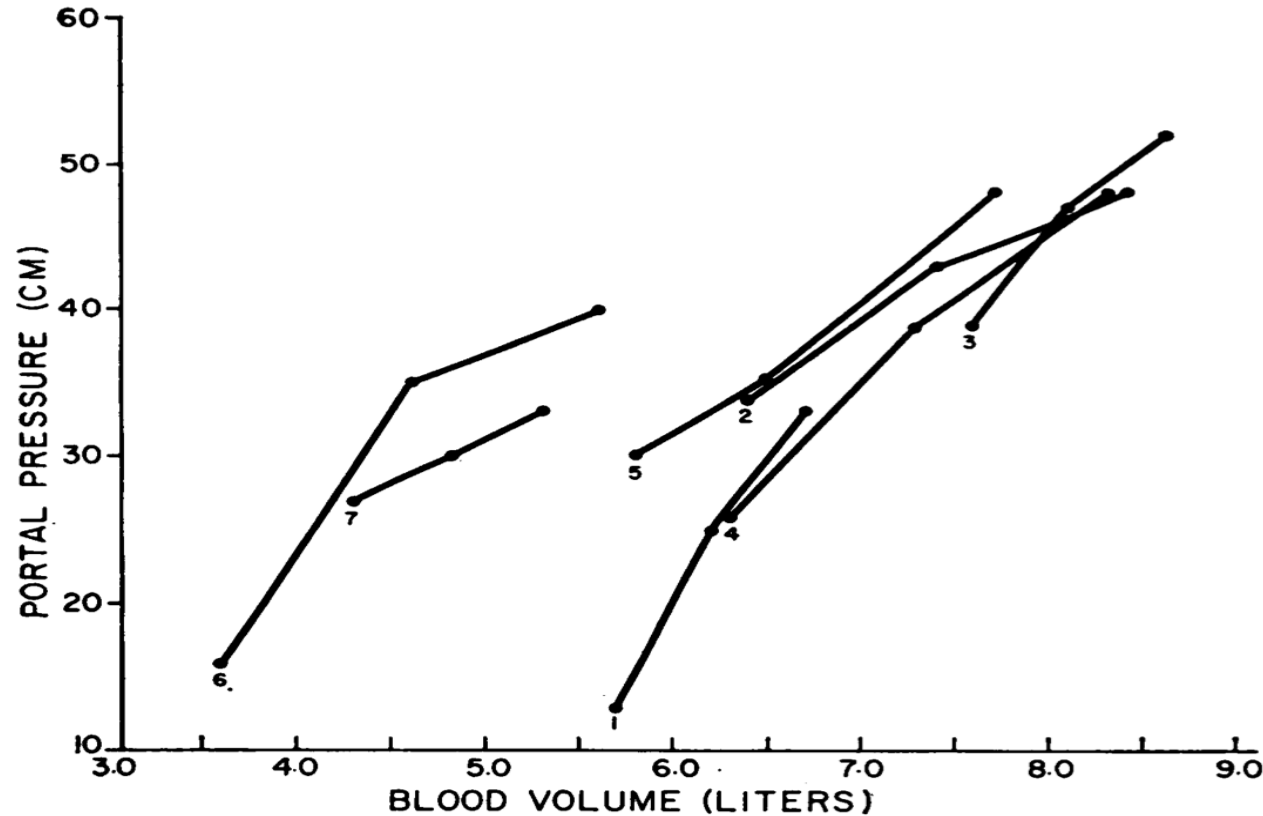
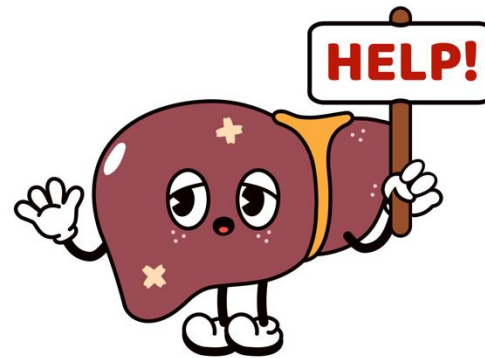


Fig. 1 *Portal pressure-blood volume curves from seven cirrhotic patients with portal hypertension derived from three measurements of portal pressure at blood volumes differing by more than 500 ml.*





TO BE CONTINUED...

